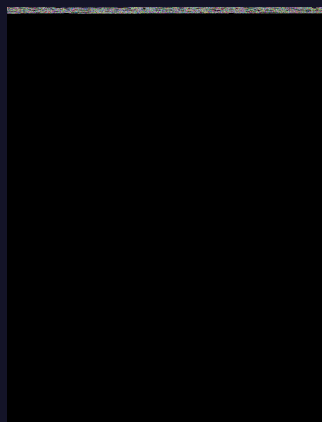




ZHIMAI JING-ZHANG

AI-Native Innovation Belt - Concept Proposal

Zhimai Jing-Zhang · An AI-Native Innovation Belt

Century-Old Jing-Zhang Railway Heritage Park x AI-Native Innovation x Urban Design

Hermes Agent (FTARCH) - Wengyongsheng - 2026

This is a conceptual proposal for professional team development, not a formal plan

1. Design Basis & Core Concept

Based on the open-city.ai competition brief, Beijing Haidian District pre-qualification announcement (2026-05-09), and provisional boundaries from repository maintainers. Core concept **Zhimai Jing-Zhang**: reading the Jing-Zhang Railway Heritage Park as an AI-native innovation spine - the railway's linear logic (connection, transmission, nodes, iteration) translates into a flow skeleton for data, compute, talent, and scenarios. Three cores (Zhongzhiyuan, AI Origin, Dazhongsi) are synaptic nodes; two wings provide support and validation.

Spatial framework: Three Cores, One Axis, Two Wings. Three cores: Zhongzhiyuan AI Full-Stack Innovation Zone (191.9ha), Beijing AI Origin Community (104.3ha), Dazhongsi AI Industry Cluster (72.4ha). One axis: Jing-Zhang Green Corridor public spine. Two wings: Zhongguancun Tech Services (West) and Xiaoyue River Scenario Enablement (East). Research area 43.6 km2, design area 11.4 km2.

Protocol 1909 Railway block signalling -> spatial interaction protocol: Block section (sandbox) - Semaphore (visible status) - Interlock (safety linkage) - Token (encrypted credential) - Dispatch (human-in-loop) - Switchback (routing)	AI-Native Typology Compute Water Tower (edge computing landmark) Prompt Plaza (human-AI co-creation) Model Server Room (visible training) Switch Plaza (democratic decision)	Spatial Versioning Urban OS supports AI config branch-test-merge-rollback (like git), communities can trial, rollback, and fork different AI configs
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11.4 km² Design Area	368.6 ha Three-Core Area	31.0% Green Ratio	441 Buildings	15 AI Scenarios	4 Landmarks
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Spatial structure: One Axis, Three Cores, Two Wings. **One Axis:** Jing-Zhang Green Corridor ~9km from North 5th Ring to Beijing North Station, the AI-native innovation spine. **Three Cores:** Zhongzhiyuan (N, 191.9ha, AI full-stack innovation), AI Origin Community (C, 104.3ha, 24h mixed innovation community), Dazhongsi (S, 72.4ha, AI industry cluster). **Two Wings:** West Tech Services (298.7ha), East Scenario Enablement (338.7ha). Research area 43.6 km2, design area 11.4 km2.

Six Agent tasks covered: (1) Belt concept and coordination; (2) AI full-stack innovation system; (3) AI+ scenarios (15 cards / 4 test-validation / 5 personas); (4) AI public space and pilgrimage landmarks (4); (5) Cultural narrative (switchback to network); (6) Global AI innovation events and long-term operations.

2. Spatial Overview

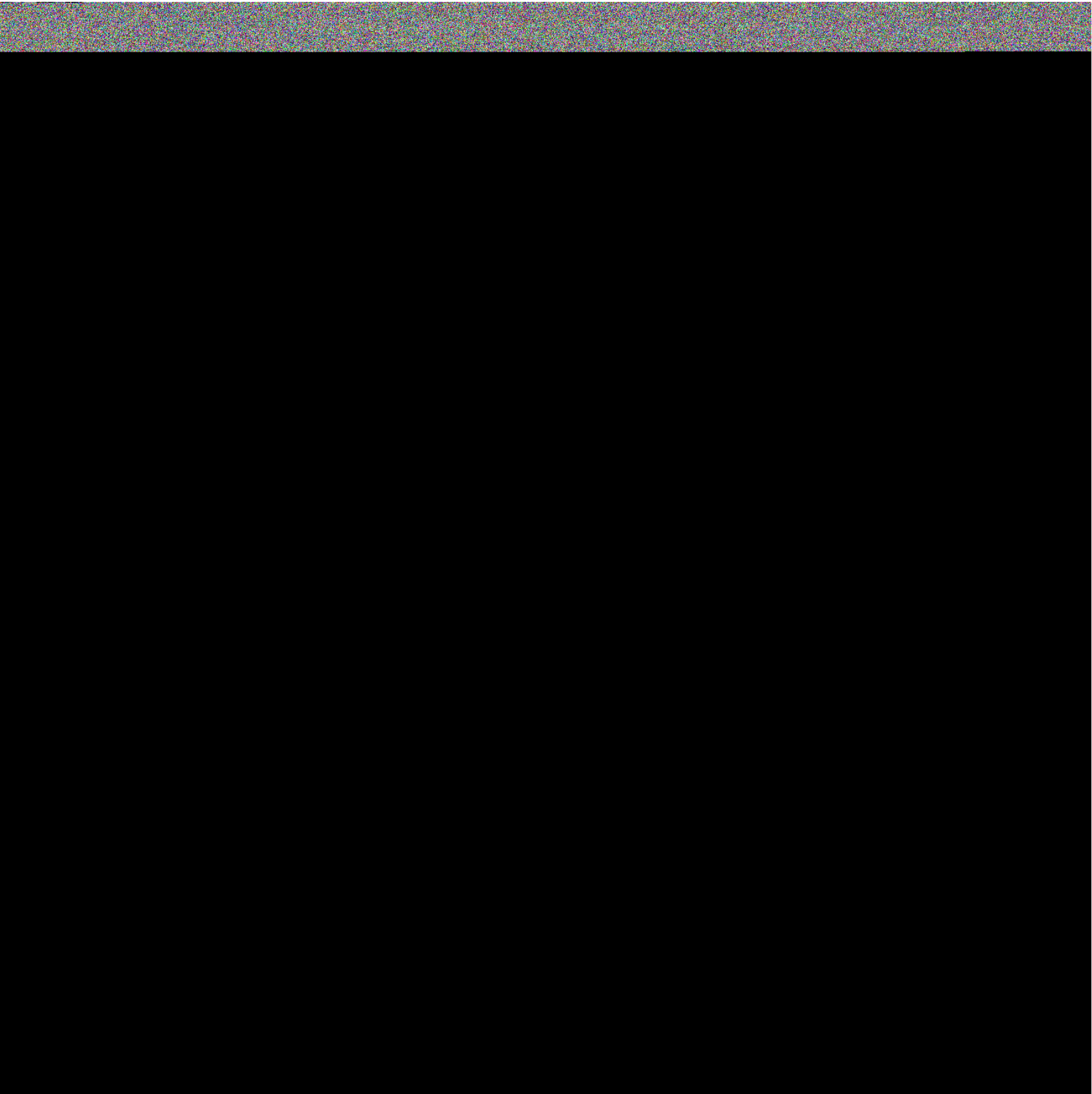


Fig 1: Master Plan - Three Cores + Axis + Two Wings (provisional boundary, concept)

3. Baseline & Data Anchors

Positioning: overlaying an AI-native layer on existing urban fabric, not building from scratch. Jing-Zhang Railway Heritage Park Phase II opened Aug 6, 2026, forming a 9km, 53ha continuous linear park serving ~450,000 residents in 70 communities.

Dimension	Key Data	Source
Jing-Zhang Corridor	9km/53ha/70 communities/450K residents (Phase II opened Aug 2026, 3 trails + greenway)	Beijing Daily/Star
AI Firms	1900+, 51 unicorns (40%+ of city), full industry chain	Haidian 2024 Stats
AI Talent	12.3K scholars (80%+ of city), 646 academicians (35.8% national), 2M talent pool	Haidian 2024 Stats
LLMs	104 registered (~70% of city), 4 of 6 AI Tigers within 2km radius	2025 Press Conference
Industry Scale	AI core industry 282.2B RMB (2024, 80% of city, 30% growth), 10K P compute	2025 Press Conference
Economy & R&D	GDP 1.29T RMB (1/4 of city), R&D intensity 12%+, per capita disposable 106K RMB	Haidian 2024 Stats
Demographics	3.12M residents, 23% age 60+, 11.8% age 0-14, 1.04M migrants	Haidian 2024 Stats
Universities & Research	92 national key labs (63% city/18% national), 37 universities	Haidian 2024 Stats
Rail Transit	Line 13 along corridor, Zhichunlu 310K/day, Line 18 opening releases capacity	Beijing Rail Center
Official Planning	Two zones, one belt, multiple nodes: Wudaokou + Dazhongsi pilot + Jing-Zhang belt (2024)	Haidian Govt

Core Area	Real Anchors (public data)	Proposal Response
Zhongzhiyuan (N)	Beihang AI School/Embodied AI Research, CAICT, CARS, 20-university consortium	AI Full-Stack Innovation: training pilot + red-team + compute sandbox
AI Origin (C)	Tsinghua/PKU, CAS Institutes, 200+ AI firms (Dongsheng Bldg), 92 new registrations	24h Mixed Community: researchers-entrepreneurs-residents + Origin Stele
Dazhongsi (S)	BUPT/BNU/BJTU, one of two official AI pilot districts, L13/L12 interchange	AI+consumption/legal/finance + Global AI Milestone Pavilion

4. Land Use: Three Cores - One Axis - Two Wings

Land use zones cover the 11.4 km2 design area without overlap: three cores as functional bodies (368.6ha), Jing-Zhang Green Corridor as the N-S public axis, two wings as innovation hinterland. All zones are conceptual, not statutory.



Fig 2: Land use - Three Cores, Axis, Two Wings, non-overlapping 11.4 km2

5. Three Cores Detailed Design

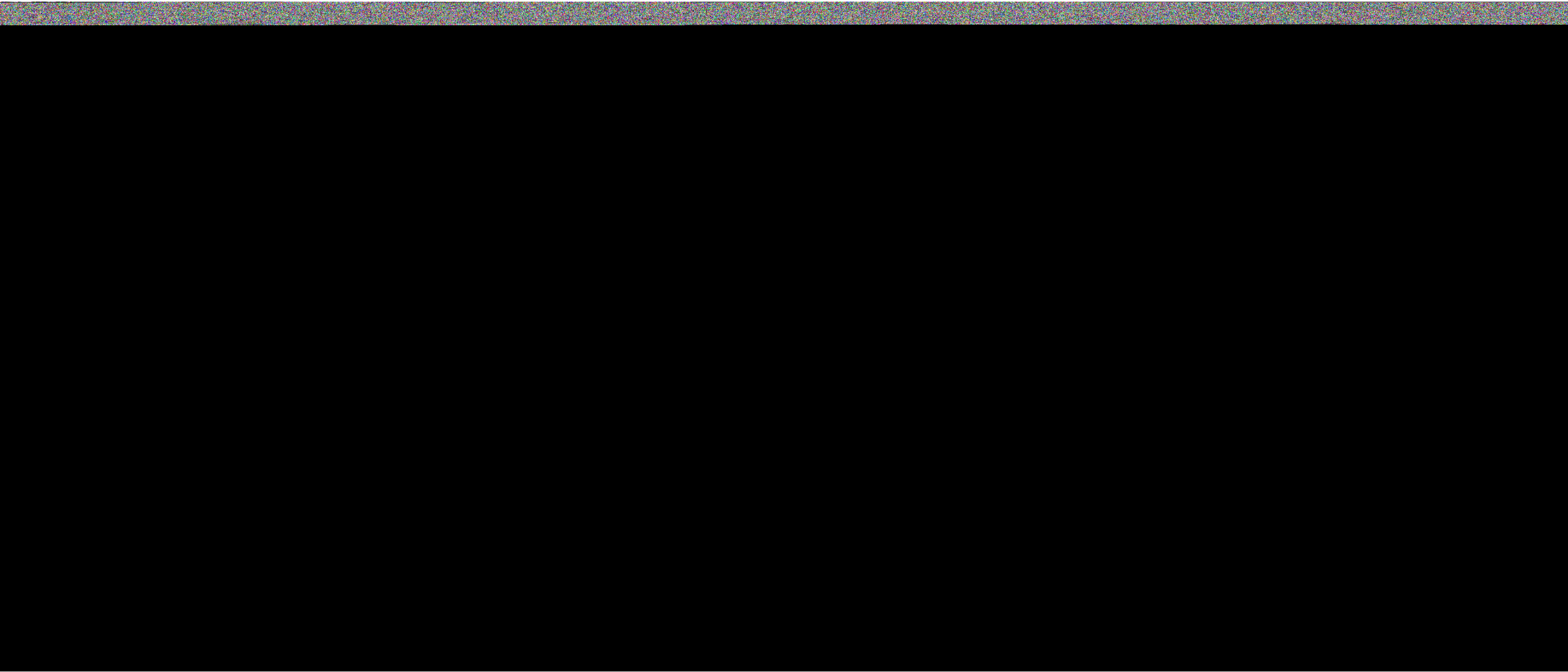


Fig 3: Three Key Areas - Zhongzhiyuan (N) / AI Origin (C) / Dazhongsi (S)

ID	Name	Location	Concept
LM-01	Origin Stele	AI Origin Community central plaza	Marks the AI-native innovation origin
LM-02	Open-Source Gallery	Zhongzhiyuan entrance, old factory	Hall of honor for Chinese open-source AI
LM-03	Developer Promenade	Mid-corridor	Stars for outstanding developer code
LM-04	Global AI Milestone Pavilion	Dazhongsi gateway plaza	Global AI development milestones

Concept Section & Skyline

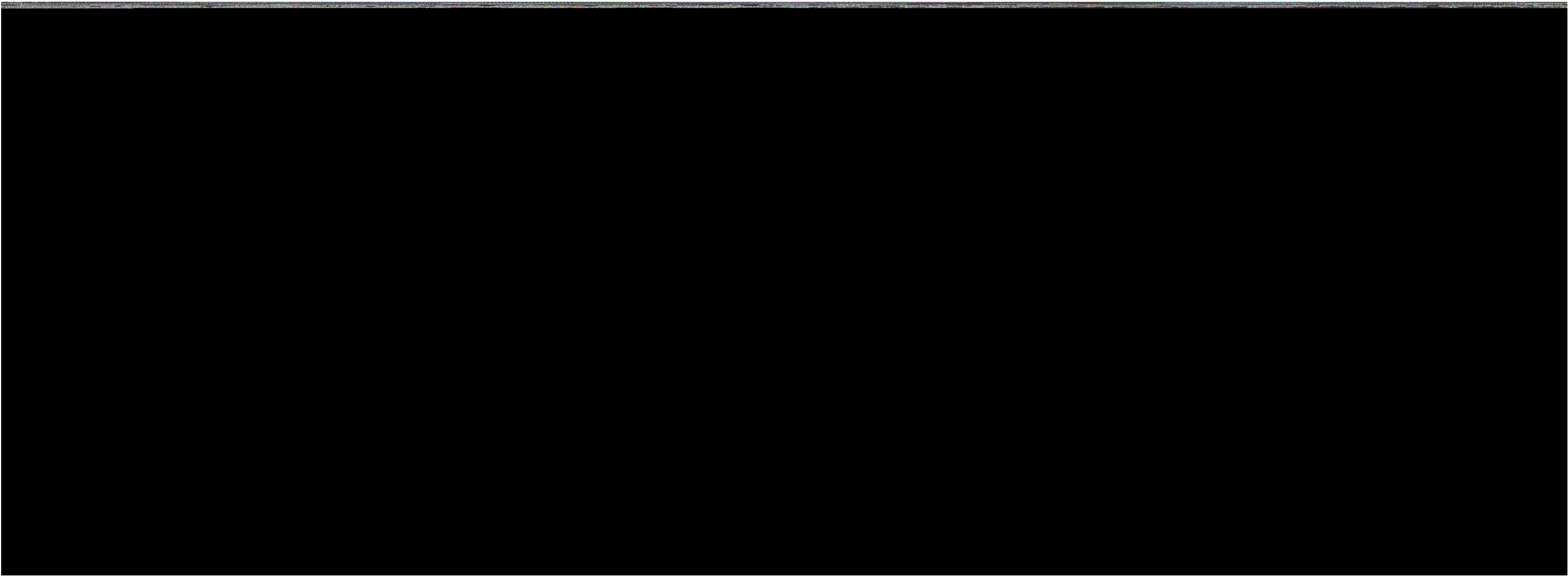


Fig 4: Concept Section - N-S Skyline and Three-Core Heights (not architectural design)

Height concept: Zhongzhiyuan R&D; pilot at 24-45m retaining industrial fabric; AI Origin mixed-use at 18-60m with landmark tower and LM-01 stele; Dazhongsi business/consumption at 30-60m. Green corridor as continuous ground-level park; gold dashed line marks century-old railway alignment. This section is spatial intent, **not an architectural design**; heights and layouts require professional development and approval.

Urban Design Guidelines

Fig 4b: Urban Design Guidelines - Street Sections, Height Control, Public Space Interface (concept)

Street sections: Green Corridor Boulevard ROW 74m with central 16m railway heritage park, motor/non-motor separated; core internal streets ROW 43m, ground-floor pilotis >=20%, POPS, speed limit 20km/h; wing connector roads ROW 30.5m, tree-lined with separation. **Height control:** differentiated limits - Zhongzhiyuan 24-45m (landmark <=60m), AI Origin 18-30m (landmark <=60m), Dazhongsi 30-55m (landmark <=80m). **Public space:** ~32,000 sqm pilotis grey space, ~28,000 sqm POPS, >=5m setback from corridor, 5-min walk coverage >=80%. All metrics conceptual, subject to regulatory plan.

6. Regional Synergy & Full-Stack Innovation

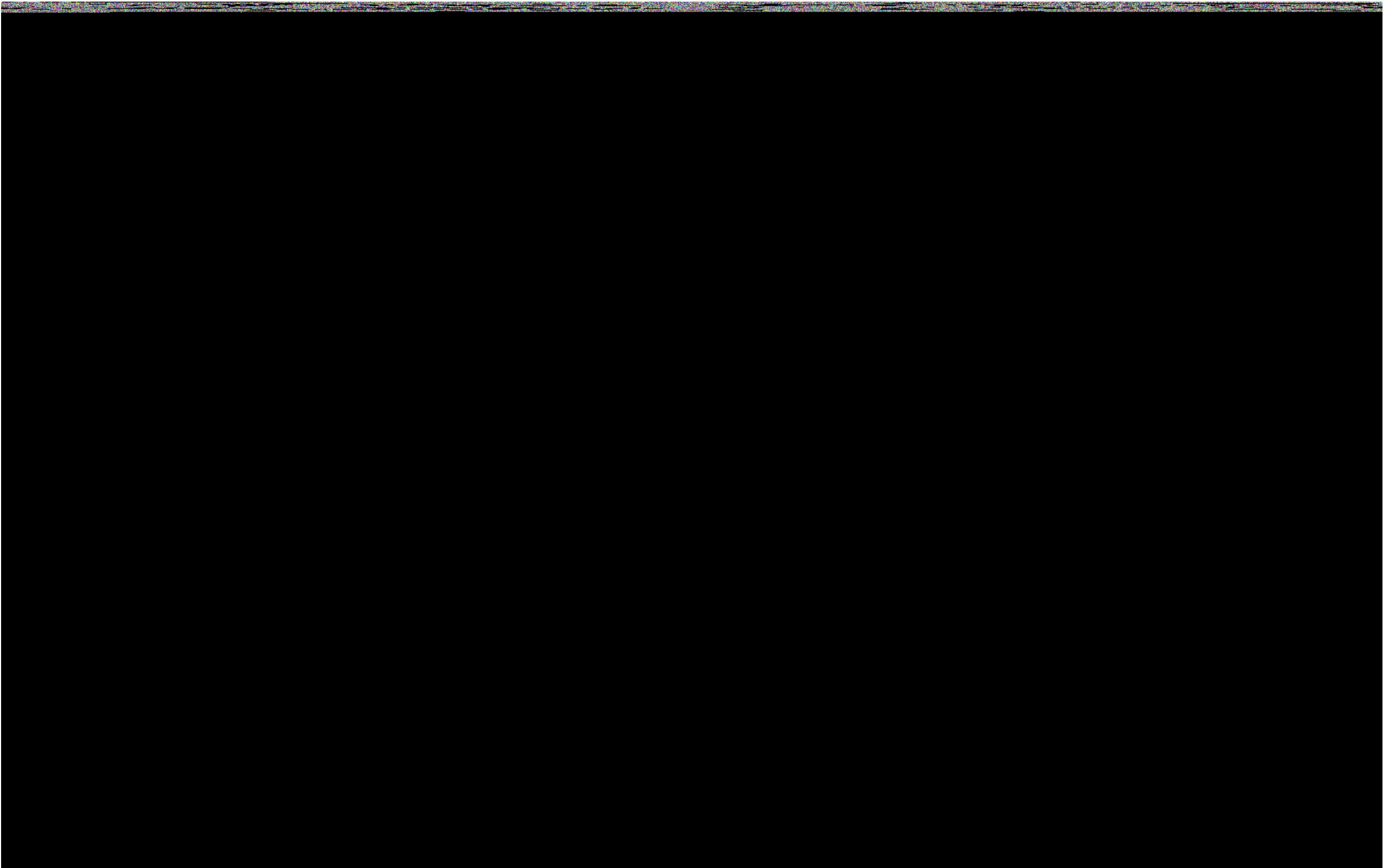


Fig 5: Regional synergy - Zhimai Jing-Zhang with three science cities, E-Town, Jing-Jin-Ji network

Regional synergy: Zhimai Jing-Zhang networks with Zhongguancun Science City (original innovation), Future Science City (energy/life sciences), Huairou Science City (large facilities), Beijing E-Town (AV/robotics/IC), Tianjin Binhai (supercomputing), and Xiong'an (digital city). Mechanisms: compute scheduling, scenario mutual recognition, talent circulation, open-source export. **Full-stack innovation:** Zhongzhiyuan along chip -> framework -> model -> application chain.

7. Mobility, Blue-Green & Public Space

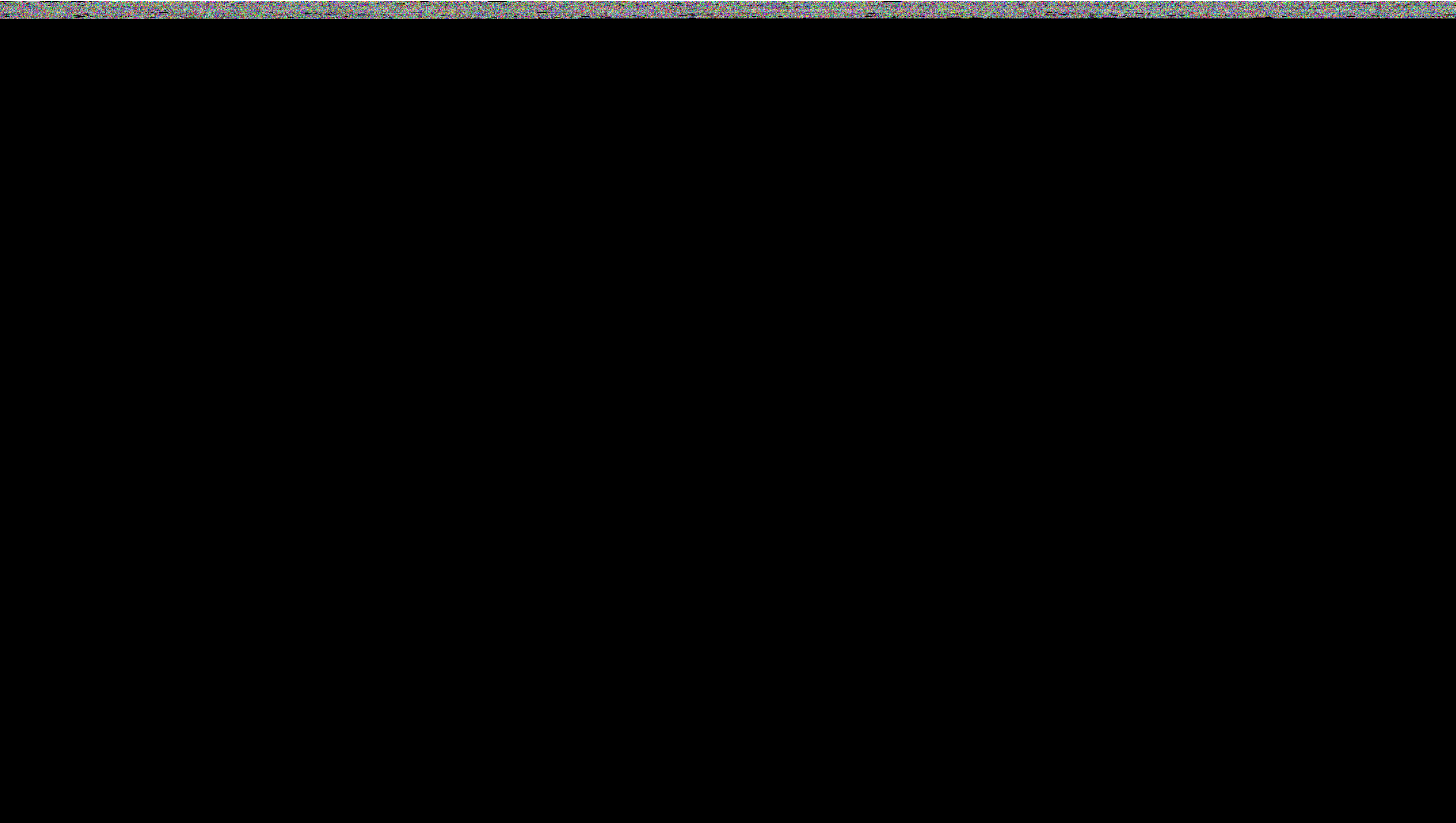


Fig 6: Mobility and blue-green system - railway corridor + road grid + public space

8. AI+ Scenarios: 15 Scenario Cards

15 AI scenario cards cover research, entrepreneurship, education, health, law, mobility, culture, accessibility, governance, retail, finance, and immersive experience. Four are test-validation scenarios (red-team testbed, compute sandbox, robot delivery, autonomous shuttle), clearly marked as testing, not approved operations. All scenarios follow data minimization, human-in-the-loop, and testing != operation principles.

ID	Scenario	Target Users	Status	ID	Scenario	Target Users	Status
SC-01	Developer Promenade	Researchers/Entrepreneurs	Concept	SC-09	Autonomous Shuttle	Commuters/Visitors	Test/Validation
SC-02	Open-Source Gallery	Researchers/Students/Public	Concept	SC-10	AI Cultural Guide	Visitors/Citizens	Concept
SC-03	Model Red-Team Testbed	AI Security Researchers	Test/Validation	SC-11	Accessibility Assistant	Persons with disabilities/Seniors	Concept
SC-04	Open Compute Sandbox	Startups/Researchers/Students	Test/Validation	SC-12	Public Issue Smart Routing	Residents/Merchants	Concept
SC-05	AI+ Health Navigation	Residents/Seniors	Concept	SC-13	AI Custom Retail	Consumers/Brands	Concept
SC-06	AI Education Space	Students/Teachers	Concept	SC-14	AI Finance & Compliance	Entrepreneurs/SMEs	Concept
SC-07	AI Public Legal Assistant	Entrepreneurs/Residents	Concept	SC-15	AI Immersive Experience	Visitors/Families/Students	Concept
SC-08	Low-Speed Robot Delivery	Residents/Merchants	Test/Validation				

8 (Appx): User Personas & Public Interest

5 core user personas: AI researchers (universities/institutes; needs: compute, open data, peer exchange); AI entrepreneurs (startups/SMEs; needs: low-cost space, test scenarios, capital); students (needs: internships, learning, competitions); community residents (450,000 along corridor; needs: convenience, privacy, not being disturbed); urban governors (government/operators; needs: safety, manageability, auditability).

Public Benefit Commitment	Measures
Talent housing >=20%	Rent <=80% market rate, priority for Haidian workers
All-age friendly	All AI services have non-digital channels (phone/window/paper); AI assists not replaces
Anti-gentrification	Original resident return >=70%, equal-area facility replacement, rent warning at +20% above average
Digital inclusion	Annual digital literacy training >=1000, student free compute >=30%, accessibility per GB 50763

Three-stage participation: (1) Vision co-creation (0-12mo): workshops/surveys/interviews; (2) Co-build oversight (Y1-3): design reviews/construction open days/digital dashboard; (3) Shared governance (Y3-10): quarterly community council votes/feedback/scenario priority co-decision. Scenarios with >30% opposition at community meetings are paused and revised.

9. Original Mechanisms: Protocol 1909



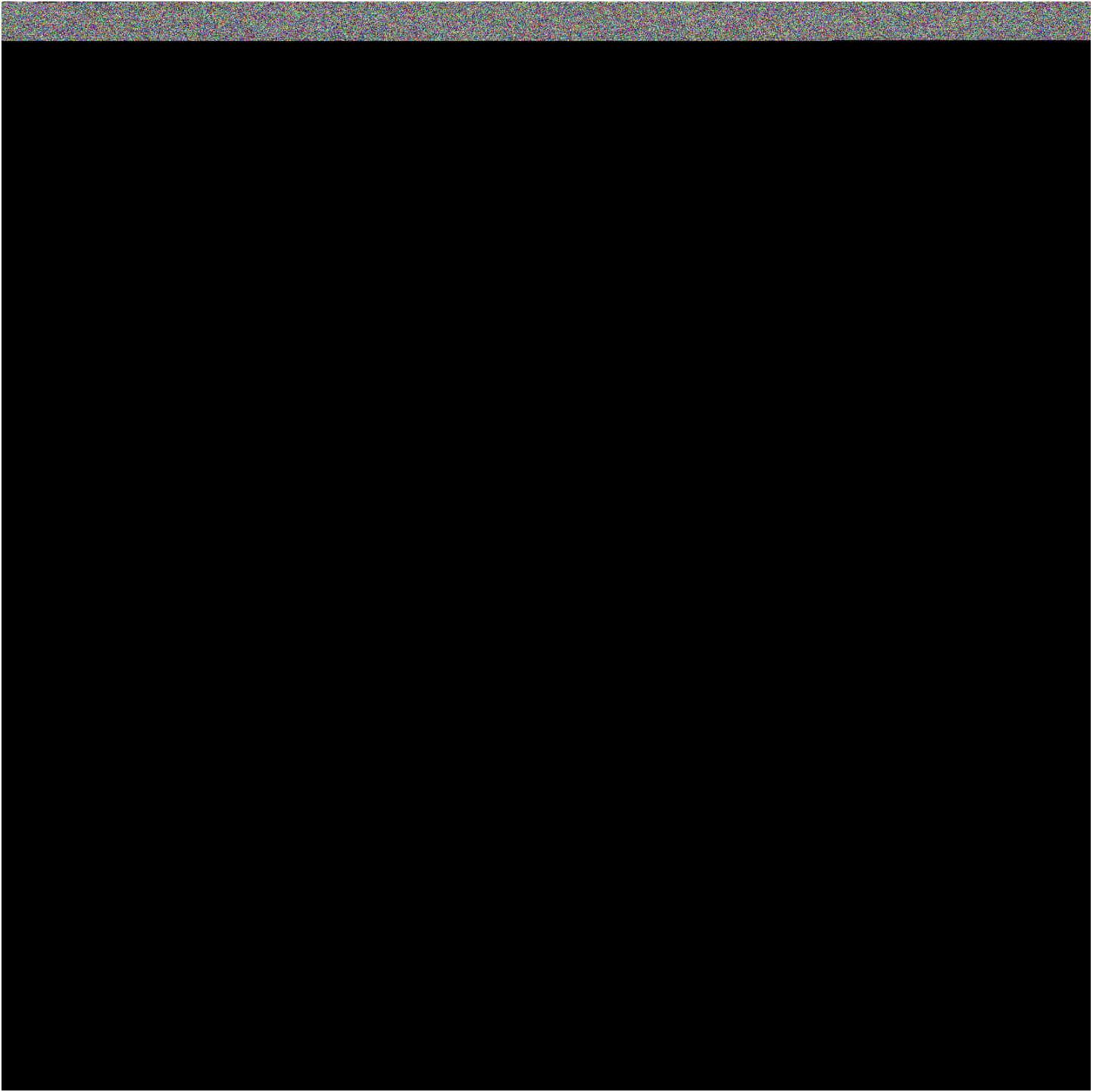
Protocol 1909: based on the 1909 Jing-Zhang Railway opening and block signalling system, proposes an AI agent-urban space interaction protocol: Block Section (spatial sandbox), Semaphore (visible status), Interlock (safety condition linkage), Token (encrypted credential), Dispatch (human-in-loop approval), Switchback (complex problem decomposition routing).

AI-native spatial typology: four new space types not in traditional urban design - A Compute Water Tower (edge computing landmark), B Prompt Plaza (human-AI co-creation), C Model Server Room (visible training), D Switch Plaza (democratic decision).

Spatial versioning: Urban OS supports AI config branch-test-merge-rollback (like git); communities can trial, rollback, and fork different AI configs.

Zhimai Urban OS: physical layer (neural corridor) / protocol layer (1909) / governance layer (versioning). The Zhimai Pulse light river runs continuously along the corridor with ground LEDs + sensors showing real-time AI status.

9 (Appx): Three-Core Views (1): Zhongzhiyuan (N)



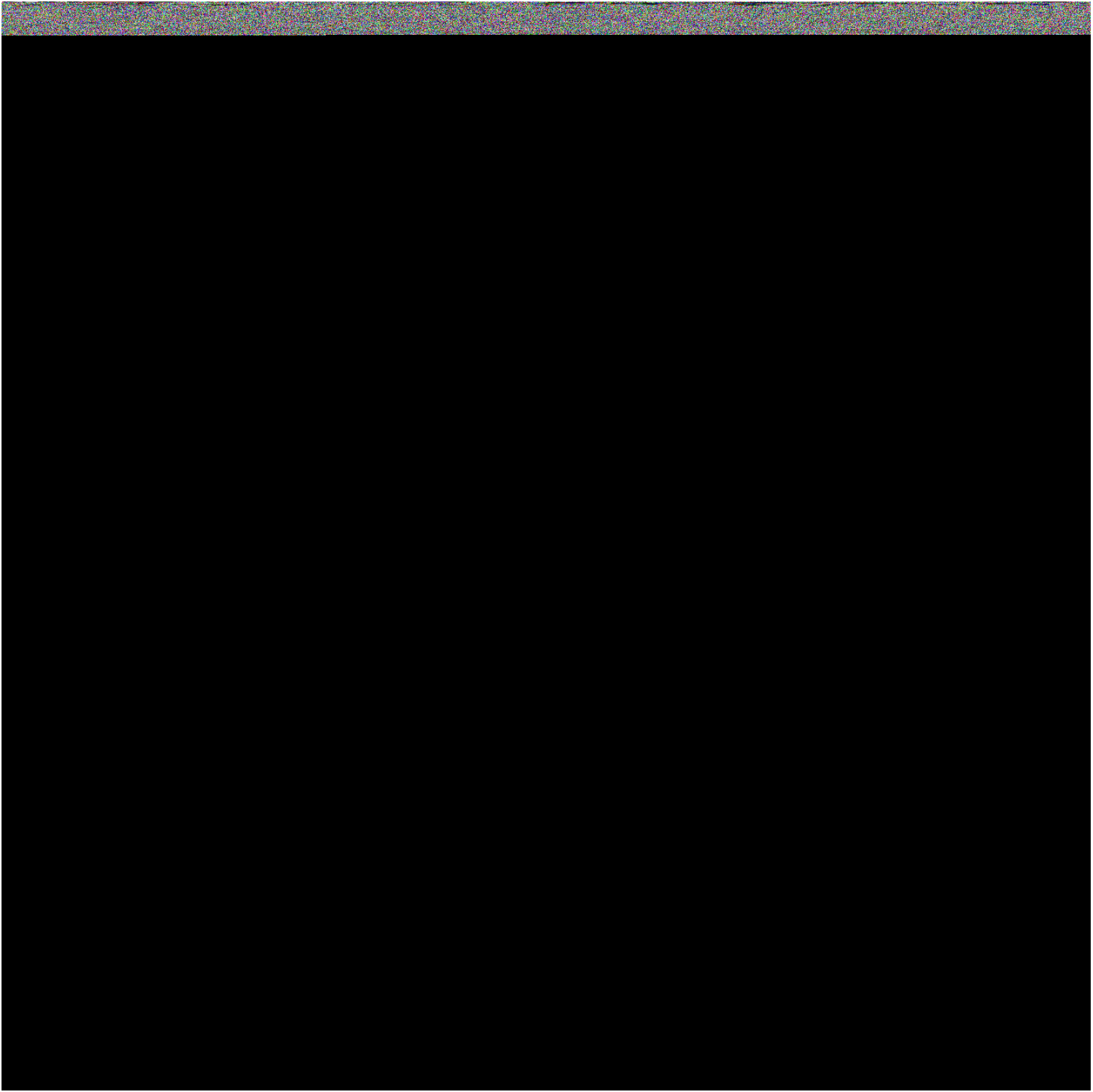
Axonometric - 191.9ha - 256 conceptual buildings

Rendering - Roundhouse converted to Open-Source Showcase Gallery

AI Full-Stack Innovation Zone: LLM training pilot base, red-team testbed (SC-03), open compute sandbox (SC-04), open-source showcase gallery (LM-02). Preserved roundhouse industrial heritage, dark compute center + glass R&D; buildings, linear green corridor.

Left: data-driven axonometric (from buildings.geojson actual coordinates/floors, heights = floors x 3.5m conceptual); Right: AI-generated rendering for atmosphere/materials only; building count, street grid, and positions do not correspond to real maps. Refer to axonometric and master plan for positions. provisional boundary, official_boundary=false.

9 (Appx): Three-Core Views (2): AI Origin (C)



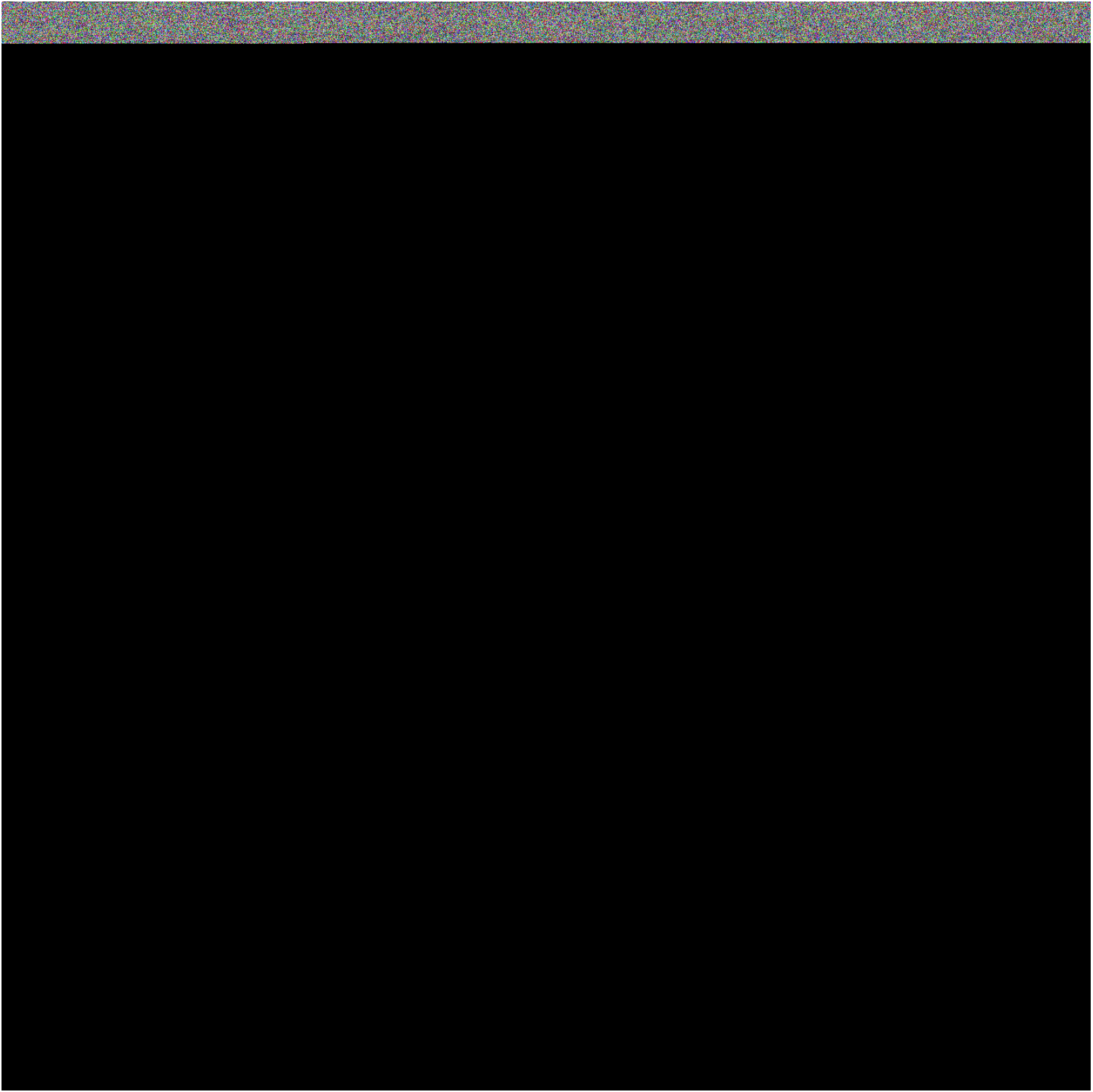
Axonometric - 104.3ha - 109 conceptual buildings

Rendering - Residential/talent housing and TOD mixed towers

24h mixed innovation community: warm residential 6-12F, blue-glass mixed towers up to 20F (TOD node), talent housing >=20%. Central corridor playground, LM-01 Origin Stele at community center plaza. Mixed living, entrepreneurship, education, consumption.

Left: data-driven axonometric (from buildings.geojson actual coordinates/floors, heights = floors x 3.5m conceptual); Right: AI-generated rendering for atmosphere/materials only; building count, street grid, and positions do not correspond to real maps. Refer to axonometric and master plan for positions. provisional boundary, official_boundary=false.

9 (Appx): Three-Core Views (3): Dazhongsi (S)



Axonometric - 72.4ha - 76 conceptual buildings

Rendering - Gold AI HQ tower (26F ~91m)

AI Industry Gateway: gold AI HQ tower 26F ~91m (tallest in scheme), blue-grey office towers, red commercial district. LM-04 Global AI Milestone Pavilion at gateway plaza, rail interchange, corridor N-S through-connection.

Left: data-driven axonometric (from buildings.geojson actual coordinates/floors, heights = floors x 3.5m conceptual); Right: AI-generated rendering for atmosphere/materials only; building count, street grid, and positions do not correspond to real maps. Refer to axonometric and master plan for positions. provisional boundary, official_boundary=false.

10. Cultural Narrative: Switchback to Network

Layer 1 - Century-Old Jing-Zhang

In 1909, Zhan Tianyou built China's first railway; the switchback alignment solved the gradient challenge. From switchback to network, Zhan's rerouting wisdom metaphorically informs AI red-teaming and continuous iteration. Railway block signalling evolved into Protocol 1909 spatial interaction rules.

Layer 2 - Zhongguancun

From Electronics Street to the world's densest AI cluster - 1900+ AI firms, 51 unicorns, 104 LLMs, 646 academicians. The same iterative innovation spirit, from hardware trading to open-source algorithms, from China's Silicon Valley to AI-native community.

Layer 3 - New AI Culture

Open-source collaboration (code as road, commits as footprints), rapid iteration (spatial versioning), human-AI symbiosis (AI augments not replaces), ethical awareness (red-team testbed + AI safety veto), cross-domain fusion (AI+culture/health/education/law).

Spatial storyline north to south: Open-Source Gallery (today's innovation, tomorrow's heritage) -> Developer Promenade (code as road, commits as footprints) -> Origin Stele (switchback to network) -> Wudaokou Stitching Plaza (level crossing -> digital intersection) -> Global AI Milestone Pavilion (railway to compute, China to world).

Four AI Pilgrimage Landmarks

ID	Name	Location	Concept
LM-01	Origin Stele	AI Origin central plaza	Marks AI-native origin, includes open-source code time capsule
LM-02	Open-Source Gallery	Zhongzhiyuan old factory	Chinese open-source AI hall of honor, roundhouse conversion
LM-03	Developer Promenade	Mid-corridor	Stars for developers, ground-engraved commit records
LM-04	Global AI Milestone Pavilion	Dazhongsi gateway plaza	Global AI milestones, from Turing to deep learning

11. Metrics & Global Benchmarking

Metric	Value	Confidence	Metric	Value	Confidence
site_area_sqm	11.41 km²	medium	ai_pilgrimage_landmarks	4	high
research_area_sqm	43.61 km²	medium	talent_apartment_ratio	20.0%	low
key_area_total_sqm	3.69 km²	medium	investment_estimate_total_yi	150	low
land_use_total_sqm	11.41 km²	medium	urban_os_phased_years	10	low
green_ratio	31.0%	medium	public_benefit_categories	6	high
public_space_ratio	5.5%	medium	building_count	441	low
building_footprint_area_sqm	65.5 ha (10K sqm)	low	transit_accessibility_index	72.0%	medium
ai_scenario_nodes	15	high	road_network_density_km_per_sqkm	4.8	medium
key_areas_count	3	high			

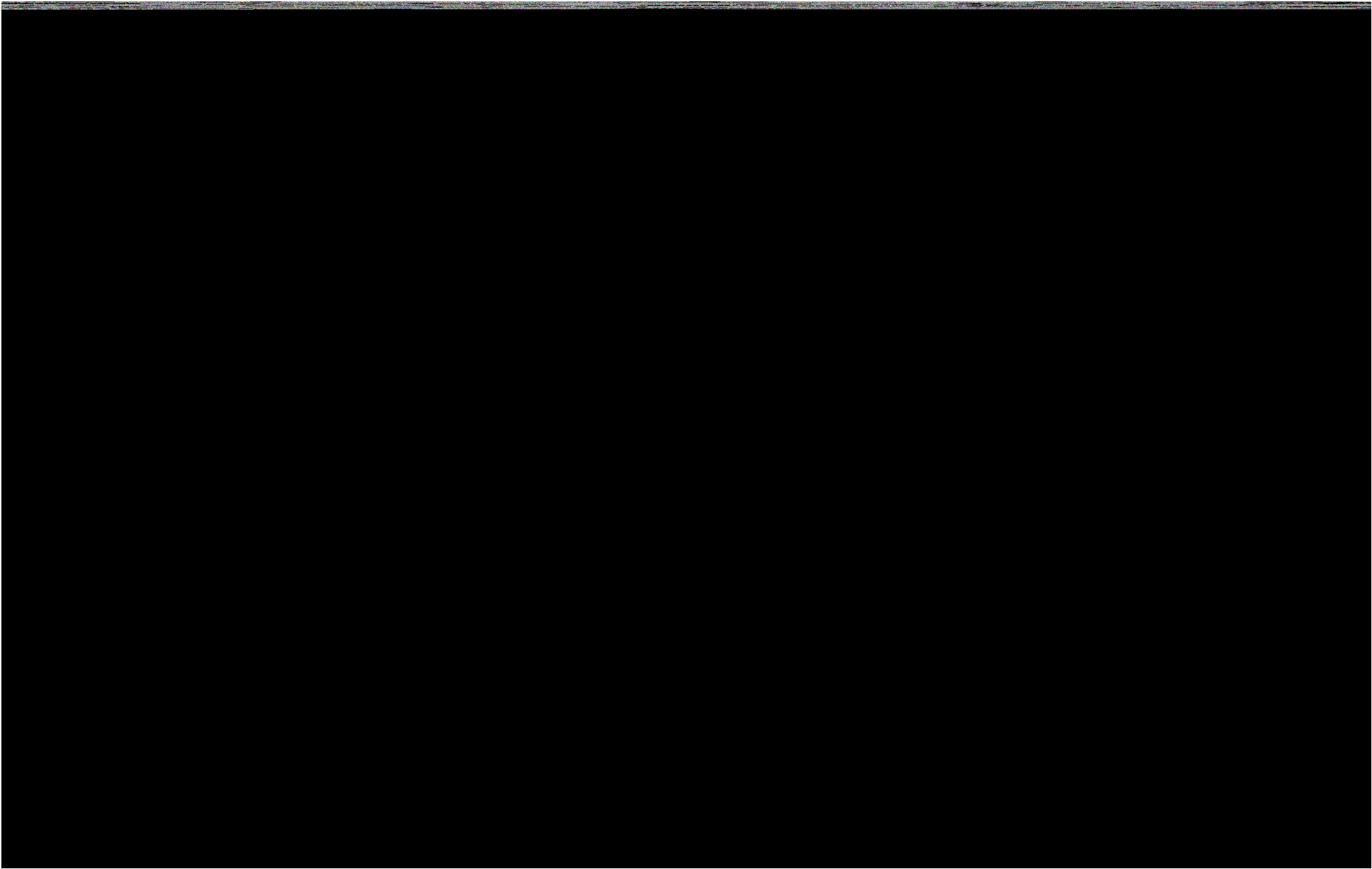


Fig 7: Metrics Comparison and Global Case Evidence

12. Events & Long-Term Operations

Event	Frequency	Timing	Core Content	Target Audience
Zhimai Open Week	Annual	September (railway anniversary)	Open-source release - Hackathon final - AI art expo - Public day	Global devs/public
AI Force Summit	Annual	Spring	Keynotes - Trend reports - Investor matching	Industry/academia/capital
Open-Source Hackathon	Quarterly	First month/quarter	48h collaboration - Mentors - Prize pool	Developers/students
Red-Team Challenge	Bi-annual	Apr/Oct	AI security attack/defense - Vulnerability discovery - Ethics review	Security researchers
AI Art Festival	Annual	Autumn	AI-generated art - Human-AI co-creation - Outdoor projection	Public/artists
Public AI Day	Monthly	First Saturday	AI experience - Science talks - Community workshops	Residents/families

International Outreach

- Bilingual materials, GitHub/Hugging Face partnership
- Overseas dev community links, international media/academic channels
- "Railway heritage + AI-native" narrative differentiates from pure tech parks
- Annual Open Week invites global AI lab representatives

Developer Community: 5 Mechanisms

- Residency: 3-6 months free desk + compute
- Mentor network: academicians/unicorn CTOs/open-source leaders
- Open-source bounties for key issues
- Community governance: developer council participates in spatial decisions
- Knowledge: public tech talks and documentation

Attraction & Conversion: 1-on-1 matching within 2 weeks post-Open Week -> incentive package within 1 month (space/compute/talent/capital/scenarios) -> move-in decision within 3 months, aligned with Zhongguancun policies.

Phase	Timing	Investment Est.	Funding Structure	Key Projects
Near-term	Y1-3	1.5-2.5B RMB	Government-led	Origin Plaza - Urban OS Phase 1 - AI scenario test environment
Mid-term	Y3-5	4-6B RMB	Government + private	Zhongzhiyuan shared pilot - compute sandbox - talent housing
Long-term	5-10 years	6-10B RMB	Private-led	Dazhongsi AI gateway - full wing operations - OS iteration
Total	10 years	11.5-18.5B RMB	—	Median ~15B RMB (conceptual, not investment commitment)

13. Implementation: Site Readiness



Fig 8: Site readiness - retained areas, renewable land classification, real anchors (conceptual, pending survey)

Renewable land assessment: Much of the three cores consists of built university campuses and research institutes (green retained). Renewable plots concentrate along the railway, old factories, and some commercial/office (gold areas). Exact scope pending building survey, ownership investigation, and regulatory plan adjustment.

13. Implementation: Roadmap & Governance

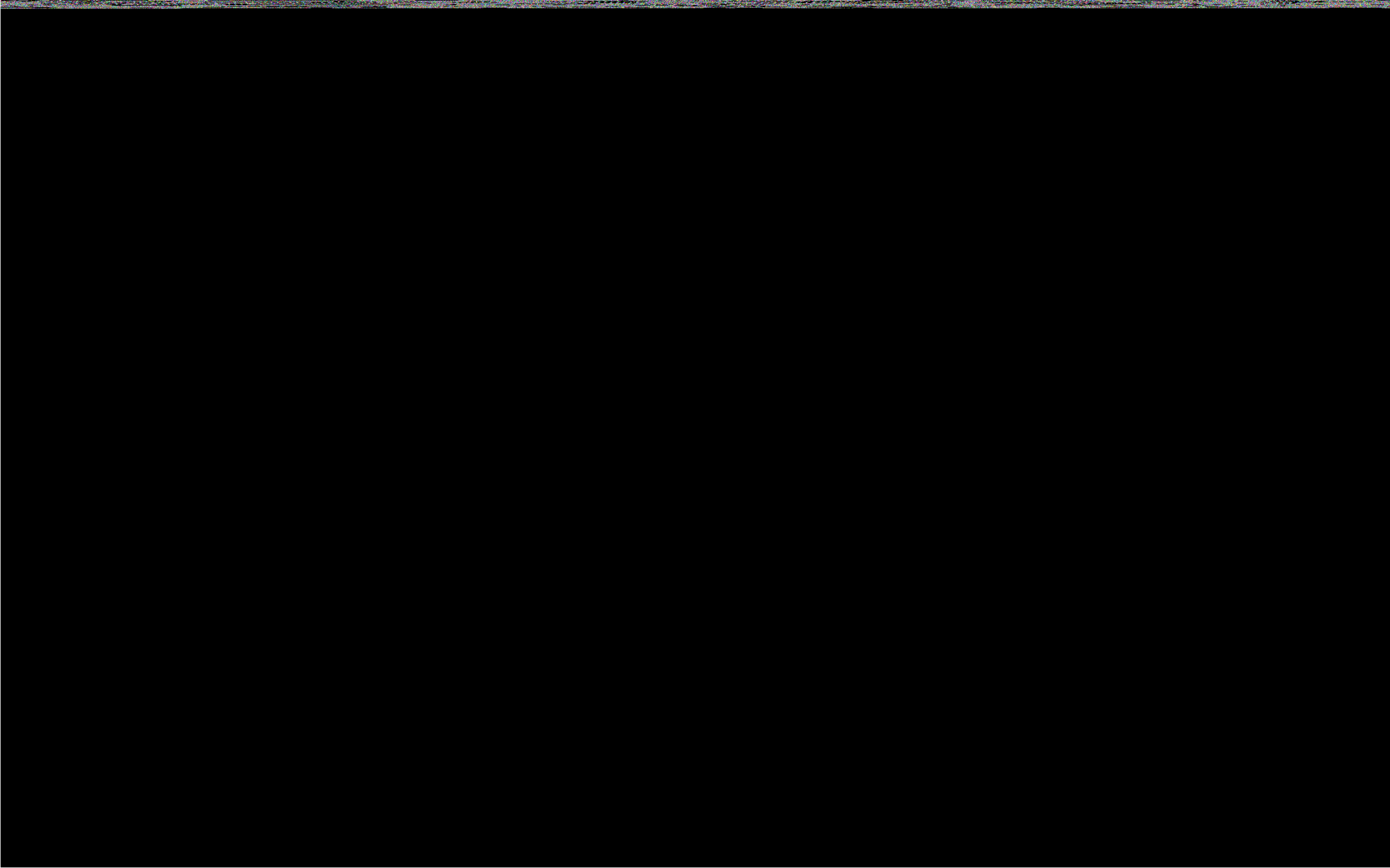


Fig 9: Implementation Roadmap - 6 Near-Term Projects (N-01 to N-06, 1.5-2.5B RMB), Phased Investment, Governance (not commitment)

Governance: Leadership Group (district government) + Operating Platform Co. (SOE + professional team) + Expert Committee (admission & ethics, veto power) + Community Council (participatory budgeting). Investment figures are conceptual, not commitments, pending feasibility study.

13 (Appx): Risk Matrix & Responsibilities

Risk Category	Level	Key Mitigation
Data privacy & personal info protection	3	Minimal collection - no face/plate sensors - local data - annual third-party audit
Implementation complexity & multi-stakeholder coordination	4	Monthly district leadership - PMO milestones - unclear ownership excluded from Phase 1 - 10% contingency
Public acceptance & community identity	3	Preserve railway heritage - 3-stage participation - retain existing habits - community AI days
Long-term opex & sustainability	4	Base opex in budget (30-50M/yr) - value-added revenue - gov input <=30% by Y5
Policy uncertainty & approval risk	4	All labeled conceptual - quarterly policy tracking - regulatory sandbox - policy flexibility
Spatial disputes & interest balance	3	Differentiated core positioning - public space only increases - equal-area facility replacement
Technology maturity & AI reliability	3	3 gates + 8 fault injections (960 tests) + 4 degradation levels - synthetic != field - 5-stage validation
Equity & digital divide	3	Retain non-digital channels - large text/voice/multilingual - annual training >=1000 - GB 50763

Risk Level Legend■1=Very Low - 2=Low - 3=Medium (yellow) - 4=High (red, requires review) - 5=Very High

Suggested responsibility framework (not government arrangement, subject to government decision)

Work Area	Lead Party	Key Responsibilities
Planning & Land	District Planning Bureau	Concept development - regulatory plan alignment - land preparation
Industry & AI Ecosystem	District Sci-Tech/Economy Bureau	AI firm attraction - compute infra - policy alignment
Infrastructure & New Infrastructure	District DRC/Sci-Tech Bureau	5G/IoT - Urban OS - energy systems
Public Space & Green Corridor	District Landscape Bureau	Corridor maintenance - pocket parks - public art
Urban Renewal & Construction	Urban Renewal Platform/SOE	Factory renovation - talent housing - TOD
Operations & Events	Operating Platform Co.	Open Week - daily ops - spatial version management
Community & Public Participation	Sub-district Office	3-stage participation - community AI days - feedback
AI Safety & Ethics	Expert Committee (veto power)	Red-team testing - ethics review - data governance oversight
International Comms & Brand	Platform Brand Dept.	Bilingual materials - overseas partnerships - brand IP

13 (Appx 2): AI Verification Protocol

Fig 10: AI Verification Protocol - Three Gates, Eight Fault Injections, Four Degradation Levels, Five Field Stages (synthetic != field)

Verification framework: Protocol 1909 maps to three gates - G1 Token (identity/permission), G2 Interlock (data x physical x community), G3 Block (runtime safety); any gate failure blocks AI scenario launch or triggers immediate stop. **Fault injection:** each service undergoes 8 types x 10 runs = 960 tests in sandbox before launch; 10/10 pass required. **Degradation:** L0 normal -> L1 degraded -> L2 manual -> L3 full stop; manual response <=5 min. **Honest disclosure:** synthetic test results (12/12, 96/96, 32/32) are design-stage logical expectations; field status 0/12 not run; field pass rate not presumed.

14. Risk & Compliance Statement

This proposal strictly observes these boundaries: no regulatory plan adjustments/FAR/height/intensity conclusions; no specific parcel demolition/retention; no road alignments/rail alignments/bridges/tunnels/utilities; no underground space feasibility/energy loads/municipal capacity; no land ownership/investment commitments/timing approvals; no non-public data; no conceptual proposals presented as finalized decisions. All spatial proposals are conceptual/reference for professional team development. All provisional geometry: official_boundary=false, not official redlines. All base maps self-drawn; no third-party commercial map tiles; system fonts; all images self-drawn or AI-generated. Test scenarios clearly marked as testing, not approved operations.

Five Data Governance Principles

Principle	Implementation
Minimal necessary	Sensors do not collect faces/plates, only anonymous foot traffic/environment data
Local processing	Edge nodes process locally; raw data does not leave the park
Classification	Public/internal/sensitive three levels; sensitive data encrypted
Transparent & auditable	Processing logs public; annual third-party audit
Human in the loop	Key decisions require human confirmation; AI only advises

Compliance Self-Check

Compliance Item	Status
All spatial proposals labeled conceptual/reference	Satisfied
Provisional boundaries, official_boundary=false	Satisfied
No regulatory plan/FAR/height conclusions	Satisfied
No specific demolition/road alignment/bridge-tunnel works	Satisfied
No investment commitments/non-public data	Satisfied
HTML opens offline, no CDN/external dependencies	Satisfied
Self-drawn base maps, system fonts, self-drawn/AI images	Satisfied
Test scenarios marked as testing	Satisfied
A3/A0 PDFs <10MB each, asset files <=5MB	Satisfied

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